

# Automated Circuit Discovery for NIDS

## Mechanistic Interpretability for SOC-Oriented Security Research

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# Presentation Outline

- 1 Motivation & Background
- 2 Technical Framework
- 3 Research Applications
- 4 SOC-Oriented Integration
- 5 Conclusion

# The Interpretability Gap in NIDS

- Modern NIDS use Deep Learning (Transformers/RNNs) for high accuracy.
- **The Black-Box Problem:** Analysts receive alerts but cannot verify the causal logic.
- **SOC Impact:** High false-positive rates and lack of trust in "AI-driven" security.
- **Solution:** Mechanistic Interpretability to reverse-engineer model internals.
- This motivates causal—not correlational—explanations.

# Interpretability Methods in NIDS: A Comparison

| Aspect                | Feature Attribution<br>(SHAP / LIME) | Rule-Based XAI      | ACDC (Proposed)                   |
|-----------------------|--------------------------------------|---------------------|-----------------------------------|
| Explanation Level     | Input features                       | Human-defined rules | Internal computational components |
| Causality             | Correlational                        | Assumed by rules    | Causal (via activation patching)  |
| Model Internals       | Ignored                              | Not applicable      | Explicitly analyzed               |
| Attack-Specific Logic | Weak                                 | Manual              | Automatically extracted           |
| SOC Usefulness        | Limited (scores only)                | High but rigid      | High and operationally actionable |
| Forensic Value        | Low                                  | Moderate            | High (circuit-level audit)        |

# Applying ACDC to NIDS: Core Idea

- ACDC is applied **post-hoc** to a trained NIDS model
- The NIDS model is treated as a computational system, not a black box
- The key objective is to identify:
  - which internal components
  - and which information pathwaysare **causally responsible** for detecting a specific attack
- This is achieved by systematically testing and pruning the model's internal computational graph

# Mapping ACDC Concepts to NIDS

| ACDC Concept        | Interpretation in NIDS                                        |
|---------------------|---------------------------------------------------------------|
| Behavior            | Specific attack type (e.g., DDoS, Scan, Brute Force)          |
| Computational Graph | NIDS model layers and feature interactions                    |
| Activation Patching | Replacing malicious traffic activations with benign baselines |
| Edge Pruning        | Removing causally irrelevant feature pathways                 |
| Circuit             | Minimal computation responsible for attack detection          |
| Faithfulness Metric | KL divergence between full and pruned model outputs           |

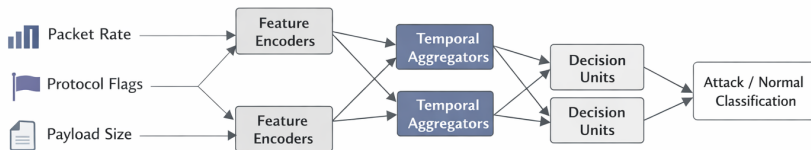
# The NIDS Computational Graph

- Represent the NIDS model as a Directed Acyclic Graph (DAG).
- **Nodes ( $V$ ):** Residual stream, Attention Heads, MLP layers.
- **Edges ( $E$ ):** Linear information flow between components (e.g., residual connections, attention outputs).
- **Task:** Prune  $E$  until only the "Security Logic" remains.

This abstraction is architecture-agnostic (Transformer, RNN, CNN).

# System Architecture: ACDC-Enabled NIDS

## NIDS as a Computational Graph





# Step A: Task and Dataset Setup

To find a circuit, we must define the behavior:

## Task Definition

Detecting SYN Flood attacks within the UNSW-NB15 dataset.

- **Clean Dataset:** Malicious traffic samples.
- **Corrupted Dataset:** Normal baseline traffic or mean-ablated features.

Behavior is quantified using the logit difference between Attack and Normal classes.

## Step B: The Pruning Criterion ( $\tau$ )

ACDC removes an edge ( $w \rightarrow v$ ) if the change in performance is below a threshold  $\tau$ .

$$D_{KL}(G \parallel H_{new}) - D_{KL}(G \parallel H) < \tau$$

- $G$ : Full NIDS Model.
- $H$ : Current Subgraph.
- $D_{KL}$ : Kullback–Leibler Divergence.

KL divergence captures distributional change, not just confidence drop.

## Step C: Activation Patching

- Patch activations from corrupted traffic into clean traffic flow.
- If prediction is unchanged, the patched edge is **not** causally required.
- This establishes causality rather than correlation.
- **Example:** replacing SYN packet-rate activations with benign statistics.

# The ACDC Algorithm

- 1: **Input:** Model  $G$ , threshold  $\tau$ , task  $T$
- 2: **Initialize:**  $H \leftarrow G$
- 3: **for** each edge  $e \in H$  (reverse topological order) **do**
- 4:     **if** metric impact  $< \tau$  **then**
- 5:          $H \leftarrow H \setminus \{e\}$
- 6:     **end if**
- 7: **end for**
- 8: **return**  $H$  (minimal security circuit)

# Research: Generalization vs. Spurious Correlation

- Does the model recognize **TCP flags** (generalization)?
- Or is it relying on specific **source identifiers** (spurious correlation)?
- ACDC exposes whether detection logic is robust or shortcut-based.
- This limitation exposes redundancy rather than model failure.

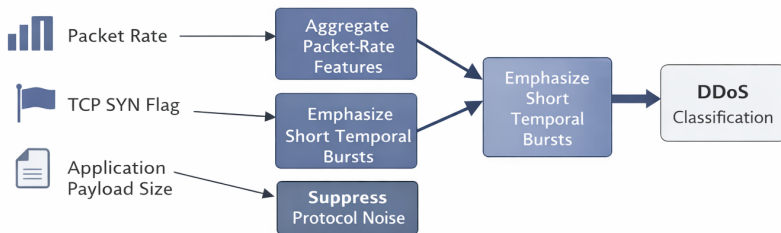
# Example: Circuit Discovery for DDoS Detection

- Consider a trained NIDS detecting volumetric DDoS attacks.
- ACDC isolates a circuit that:
  - aggregates packet-rate features,
  - emphasizes short-term temporal bursts,
  - suppresses protocol-specific noise.
- The circuit explains *how* the model detects flooding behavior.

# ACDC-Based Circuit Extraction for DDoS Attacks

## Example: Circuit Discovery for DDoS Detection

ACDC isolates a DDoS-specific circuit within the trained NIDS



# Research: Adversarial Attack Resilience

- Adversarial traffic often induces **pathway divergence**, activating circuits normally used for benign traffic.
- Research Question: To what extent can adversarial perturbations reroute computation away from the attack circuit while normal-traffic pathways remain active?
- ACDC enables circuit-level robustness analysis.



# Technical Challenge: Scalability

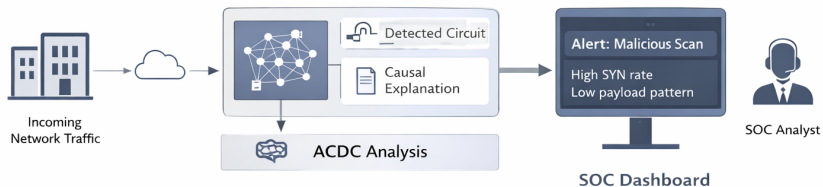
- Activation patching is computationally expensive for high-throughput traffic.
- **Research Focus:** Attribution patching to approximate circuits efficiently.
- ACDC is intended for offline analysis, not inline detection.

# Failure Case: The OR-Gate Problem

- Redundant detection pathways may exist for the same attack.
- ACDC may prune both if individual removals do not reduce performance.
- This remains an open research challenge.

# System Design: ACDC-Enabled NIDS

## SOC-oriented Explainable Alerts



# SOC Integration: Transparent Alerts

Instead of: *Alert: Anomaly detected*

## ACDC-Enhanced Alert

Logic: Observed high SYN count combined with low data payload patterns.

Causal confidence: 99.2%.

Circuits are summarized, not shown raw, to analysts.

# Automated Forensics for Tier-3 Analysts

- ACDC extracts the circuit responsible for new attack detections.
- Analysts can inspect the exact behavioral logic used by the model.
- Enables faster root-cause analysis for advanced threats.

# Logic-Based Filter Tuning and Evaluation

- Circuit inspection identifies causes of false positives.
- Model refinement is applied only to implicated sub-components.
- Evaluation metrics:
  - KL divergence under ablation,
  - detection accuracy preservation,
  - circuit sparsity.

# Building Operational Trust

- ACDC provides an auditable explanation for each alert.
- Supports compliance in regulated environments.
- Enhances analyst trust in AI-driven security systems.

- Framed Network Intrusion Detection as a **mechanistic interpretability problem** rather than a black-box prediction task.
- Applied Automatic Circuit Discovery (ACDC) to extract **causal, attack-specific computational circuits** from trained NIDS models.
- Demonstrated how circuit-level explanations enable **auditable, trust-oriented decision support** for SOC operations.



- Scalable and approximate ACDC methods for high-speed network environments.
- Circuit discovery under redundancy and overlapping attack logic.
- Circuit-to-rule translation for deployment in SOC workflows.
- Behavioral analysis of zero-day attacks using circuit-level explanations.
- Human-in-the-loop refinement of interpretability for trusted NIDS.

Thank You!